

Analysis I

OLIVER WEST

Contents

1	Numerical sequences and series	1
1.1	Basics	1
1.2	Bolzano-Weierstrass theorem	4
1.3	Cauchy sequences	5
1.4	Series and convergent series	6
2	Continuity of functions	14
2.1	Limits of functions	14
2.2	Continuity of functions	16
2.3	Extreme value theorem	18
2.4	Intermediate value theorem	19
3	Differentiation	21
3.1	Basics	21
3.2	Mean value theorems	25
3.3	Higher derivatives and Taylor's theorem	29
4	Integration of functions	34
4.1	Basics	34

Lecture 1 – 22/01/26

Analysis is about things that are small, things that are large, and quantify how small or large they are.

§1 Numerical sequences and series

§1.1 Basics

Definition 1.1. A sequence $(x_n)_{n \in \mathbb{N}}$ on a set X is an enumerated list where each element is in X .

In this chapter, $X \subseteq \mathbb{R}$ or $X \subseteq \mathbb{C}$.

What does it mean for a sequence (x_n) to converge to a point $x \in X$? What does it mean for (x_n) to diverge to infinity?

For convergence, we want $|x_n - x|$ to be small. But what is small? In fact, we want $|x_n - x|$ to be arbitrarily small, i.e. smaller than any $\varepsilon > 0$ we choose. What really

matters is the *tail* of the sequence – we can always ignore the first N terms, and we should pick N depending on ε .

For divergence to infinity, we want $|x_n|$ to get arbitrarily large, i.e. larger than any $L > 0$ we choose. Again, we ignore the first N terms depending on L .

Definition 1.2. We say (x_n) converges to some finite x , $x_n \rightarrow x$, if, for any $\varepsilon > 0$, there exists $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$.

We say (x_n) diverges to infinity if, for any $L > 0$, there exists $N(L)$ such that $x_n > L$ for all $n \geq N$, analogously for negative infinity. For complex sequences, we must consider $|x_n|$.

Example 1.3 • Take $x_n = \frac{1}{n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{n} < \varepsilon$ for any $n \geq N = 1 + \lceil \frac{1}{\varepsilon} \rceil$.

• Take $x_n = \frac{1}{2^n}$, then $x_n \rightarrow 0$ because $|x_n - 0| = \frac{1}{2^n} < \varepsilon$ for any $n \geq N = \max(1 + \lceil \log_2(\frac{1}{\varepsilon}) \rceil, 1)$.

• Take $x_n = in$, then $x_n \rightarrow \infty$, since $\frac{x_n}{i} = n > L$ for any $n \geq N = 1 + \lceil L \rceil$.

• Take $x_n = (-1)^n$, then x_n doesn't converge or diverge to infinity.

Lemma 1.4

If it exists, the limit of (x_n) is unique.

Proof. Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Take $\varepsilon > 0$, then we have

$$\exists N_1 = N_1(\varepsilon) \text{ s.t. } |x_n - a| < \varepsilon \forall n \geq N_1$$

$$\exists N_2 = N_2(\varepsilon) \text{ s.t. } |x_n - b| < \varepsilon \forall n \geq N_2$$

Now take $N = \max(N_1, N_2)$. We have

$$\begin{aligned} |a - b| &= |a - x_n + x_n - b| \\ &\leq |x_n - a| + |x_n - b| \\ &< 2\varepsilon \end{aligned}$$

for all $\varepsilon > 0$. It follows that $|a - b| = 0$, so $a = b$. ■

This justifies the notation $\lim_{n \rightarrow \infty} x_n = x$.

Proposition 1.5 (Sandwich theorems)

Let $(x_n), (y_n), (z_n)$ be real sequences. Then

- If $x_n \leq y$ for all n and $x_n \rightarrow x$, then $x \leq y$. Note that $x_n < y$ does not imply $x < y$.
- If $x_n \rightarrow x$, $z_n \rightarrow x$ and $x_n \leq y_n \leq z_n$ for all n , then $y_n \rightarrow x$.

Proof. Left as an exercise. ■

Lemma 1.6

Let (x_n) be a complex sequence. Then $x_n \rightarrow x$ if and only if $\operatorname{Re}(x_n) \rightarrow \operatorname{Re}(x)$ and $\operatorname{Im}(x_n) \rightarrow \operatorname{Im}(x)$.

Proof. Recall that, for $z \in \mathbb{C}$, $|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$.

$\Rightarrow$ Use $|\operatorname{Re}(z)| \leq |z|$ and $|\operatorname{Im}(z)| \leq |z|$ with $z = x_n - x$.

$\Leftarrow$ Use $|z| \leq |\operatorname{Re}(z)| + |\operatorname{Im}(z)|$, again with $z = x_n - x$. Fixing $\varepsilon > 0$, we can get $|x_n - x| < 2\varepsilon$, so we're done. ■

Lemma 1.7

Let $x_n \rightarrow x$, $y_n \rightarrow y$. Then $x_n + y_n \rightarrow x + y$, $x_n y_n \rightarrow xy$.

Proof. The first part is analogous to the above result on complex sequences and is an exercise.

For the second part, we have

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x y_n + x y_n - xy| \\ &\leq |y_n| |x_n - x| + |x| |y_n - y| \end{aligned}$$

Now, for $\varepsilon > 0$, we can pick an N to make $|x_n - x| < \varepsilon$, $|y_n - y| < \varepsilon$. Hence

$$\begin{aligned} |x_n y_n - xy| &< \varepsilon(|x| + |y_n|) \\ &< \varepsilon(|x| + Y) \end{aligned} \quad \text{by lemma 1.9, } Y > 0$$

and so picking $\varepsilon = \frac{\varepsilon'}{|x| + Y}$ shows the result. ■

Definition 1.8 (Bounded sequences). We say (x_n) is *bounded* if there exists $M > 0$ such that $|x_n| \leq M$ for all n , or, equivalently, if $\sup |x_n| \leq M$.

Lemma 1.9

Convergent sequences are bounded.

Proof. Take $\varepsilon = 1$, then there exists N such that $|x_n - x| \leq 1$ for any $n \geq N$, and so $|x_n| \leq 1 + |x|$ for any $n \geq N$. But for $n < N$, $|x_n| \leq \max_{1 \leq i < N} |x_i| < \infty$. ■

Definition 1.10 (Monotone sequences). We say a real sequence (x_n) is *monotone* if either

- it is increasing;
- or it is decreasing.

Proposition 1.11 (Monotone convergence theorem)

Every bounded monotone sequence converges.

Proof sketch. Without loss of generality, suppose the sequence is strictly increasing. Using the least upper bound property, we can show that $x_n \rightarrow \sup\{x_n\}$. ■

Can we strengthen to non-monotonic sequences? Clearly not fully, since $x_n = (-1)^n$ is bounded but not convergent. But, if we delete all the even terms, then it indeed becomes convergent.

§1.2 Bolzano-Weierstrass theorem**Theorem 1.12** (Bolzano-Weierstrass)

If (x_n) is a real bounded sequence, then it has at least one convergent subsequence.

Definition 1.13 (Subsequences). Given a sequence (x_n) , a subsequence of (x_n) is a sequence of the form $(x_{n_k})_{k \in \mathbb{N}}$, where (n_k) is a strictly increasing sequence of natural numbers.

Lemma 1.14

If $x_n \rightarrow x$, then any subsequence (x_{n_k}) of (x_n) must converge to x .

Proof. Since $n_k < n_{k+1}$, $n_{k+1} \geq n_k + 1$. By induction, we can show $n_k \geq k$.

Now, take $\varepsilon > 0$, so we have $N(\varepsilon)$ such that $|x_n - x| < \varepsilon$ for any $n \geq N$. So, if $k \geq N$, then $n_k \geq k \geq N$, and so $|x_{n_k} - x| < \varepsilon$, so $\lim_{k \rightarrow \infty} x_{n_k} = x$ as required. ■

Proposition 1.15 (Nested interval property)

Let $(I_n)_{n \in \mathbb{N}}$ be a sequence of closed intervals in $\mathbb{R}$ satisfying $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, and suppose $I_n = [a_n, b_n]$. If $b_n - a_n = |I_n| \rightarrow 0$ as $n \rightarrow \infty$, then $\bigcap_{n \in \mathbb{N}} I_n$ contains exactly one point.

Proof. We can see that (a_n) and (b_n) are bounded monotone sequences, so we probably want to apply [proposition 1.11](#). Indeed, we have $a_n \rightarrow a$, $b_n \rightarrow b$. Since limits preserve inequalities and $a_n \leq b_n$, we have $a \leq b$. Now we are ready to show the result.

- *Existence.* For all $k \geq n$, $a_k \in [a_k, b_k] \subseteq [a_n, b_n]$, and so $a_n \leq a_k \leq b_n$. Taking k to infinity, we have $a_n \leq a \leq b_n$ for every n , so $a \in I_n$ for every n , so it is in their intersection.
- *Uniqueness.* We know $b_n - a_n \rightarrow b - a$. But we know $b_n - a_n \rightarrow 0$, so $a = b$. Hence, if x is in the intersection, $a_n \leq x \leq b_n$ for all n , so $a \leq x \leq a$, and hence $x = a$ as required. ■

Proof of theorem. We are given (x_n) and $M > 0$ such that $|x_n| \leq M$ for any n . We will construct a series of nested intervals from which we can sample our subsequence and apply [proposition 1.15](#).

Let $a_1 = -M$, $b_1 = M$, so $I_1 = [-M, M] \supset (x_n)$. Now take $c = \frac{a_1+b_1}{2}$. Then, at least one of the halves $[a_1, c]$ and $[c, b_1]$ will have infinitely many elements of (x_n) – we take I_2 to be a half that has infinitely many elements. Continue inductively to get (I_n) , which are nested by construction and all of which have infinitely many elements of the sequence – and by [proposition 1.15](#), there exists exactly one x in their intersection.

We choose (x_{n_k}) as follows: pick n_1 such that $x_{n_1} \in I_1$. Then I_2 has infinitely many elements of (x_n) with index greater than n_1 , so pick $n_2 > n_1$ such that $x_{n_2} \in I_2$. Continue inductively.

We have $x_{n_k} \in I_k$ for every k , so $x_{n_k} \in \bigcap_{n \leq k} I_n$, and so $x_{n_k} \rightarrow x$. ■

Corollary 1.16

The Bolzano-Weierstrass theorem holds in $\mathbb{C}$.

§1.3 Cauchy sequences

A Cauchy sequence is a sequence in which the elements are getting closer as the sequence progresses.

Definition 1.17. A sequence (x_n) on $X \subseteq \mathbb{C}$ is *Cauchy* if, for any $\varepsilon > 0$, there exists N such that, for all $n, m \geq N$, $|x_n - x_m| < \varepsilon$.

Example • Take $x_n = \frac{1}{n}$. Assume wlog $m \geq n$. Then

$$\left| \frac{1}{n} - \frac{1}{m} \right| = \frac{1}{n} - \frac{1}{m} = \frac{m-n}{mn} \leq \frac{1}{n} < \varepsilon$$

and this holds for any $m \geq n \geq N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$. So (x_n) is Cauchy.

- Take $x_n = (-1)^n$. This is not a Cauchy sequence – take $n = 2k$, $m = 2k + 1$ for any $k \in \mathbb{N}$, so $|x_n - x_m| = 2$. Hence we can't take $\varepsilon < 2$.
- Take (x_n) on $\mathbb{Q}$ defined by truncation of the decimal expansion of $\sqrt{2}$. Assume wlog $m \geq n$, so

$$|x_n - x_m| < 10^{-(n-1)} \rightarrow 0$$

Note that this sequence does not converge in $\mathbb{Q}$.

Lemma 1.18

If (x_n) is Cauchy, then it is bounded.

Proof. Take $\varepsilon = 1$. Then we have N such that

$$|x_n - x_N| < 1$$

for any $n \geq N$ (we have fixed $m = N$). So $|x_n| < 1 + |x_N|$. ■

Lemma 1.19

A complex sequence (x_n) is Cauchy if and only if $(\operatorname{Re} x_n)$ and $(\operatorname{Im} x_n)$ are both Cauchy.

Lemma 1.20

If $x_n \rightarrow x$, then (x_n) is Cauchy.

Proof. For any $\varepsilon > 0$, we have N such that

$$\begin{aligned} |x_n - x_m| &\leq |x_n - x| + |x_m - x| \\ &\leq 2\varepsilon \end{aligned}$$

for $n, m \geq N$. This completes the proof. ■

Is the converse true? Not in $\mathbb{Q}$ at least – see $\sqrt{2}$ example. But we can still aspire to the converse being true in $\mathbb{R}$.

Theorem 1.21 (Completeness of $\mathbb{R}$ and $\mathbb{C}$)

Every Cauchy sequence in $\mathbb{R}$ or $\mathbb{C}$ converges.

Proof. Suppose (x_n) is Cauchy. Then it is bounded, and so by Bolzano-Weierstrass it has a convergent subsequence (x_{n_k}) converging to some x . We claim $x_n \rightarrow x$. Observe

$$|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x|$$

Taking $\varepsilon > 0$, Cauchy gives $|x_n - x_{n_k}| < \varepsilon$ for $n, n_k \geq N_1$, and $x_{n_k} \rightarrow x$ gives $|x_{n_k} - x| < \varepsilon$ for $k \geq N_2$. Fixing $k \geq N_2$ such that $n_k \geq N_1$, we have $|x_n - x| < 2\varepsilon$ for every $n \geq N_1$, which suffices for the result. ■

Remark. This is nice – it's our first way to prove convergence without having to guess the limit.

§1.4 Series and convergent series

Definition 1.22. Let (a_n) be a sequence in $\mathbb{R}$ or $\mathbb{C}$. We say that $\sum_{n=1}^{\infty} a_n$ is a *series*, and we say it converges if the sequence of partial sums $s_k = \sum_{n=1}^k a_n$ converges to some finite s . In this case, s is called the sum of the series, and we write

$$s = \sum_{n=1}^{\infty} a_n$$

Example • $\sum_{n=1}^{\infty} n$ does not converge: $s_k = \frac{n(n+1)}{2}$.

- geometric series: $\sum_{n=1}^{\infty} r^n$ converges iff $|r| < 1$. We have $s_k = \frac{1-r^{k+1}}{1-r}$ (unless $r = 1$).
- $\sum_{n=1}^i n \text{fty} \frac{1}{n(n+1)}$ converges to 1. We have

$$s_k = \sum_{n=1}^k \frac{1}{n(n+1)} = \sum_{n=1}^k \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{k+1} \rightarrow 1$$

Lemma 1.23

Fix $\lambda \in \mathbb{C}$. If $\sum a_n$ and $\sum b_n$ converge, then $\sum(\lambda a_n + b_n)$ also converges

Proof. Exercise. ■

As usual, we should understand that the tail of the series is what is significant to its convergence.

Lemma 1.24

If $a_n = b_n$ for all $n \geq N$, then $\sum a_n$ converges if and only if $\sum b_n$ converges.

Proof. We have

$$\begin{aligned} r_k &= \sum_{n=1}^k b_n \\ &= \sum_{n=1}^{N-1} b_n + \sum_{n=N}^k a_n \\ &= \underbrace{\sum_{n=1}^{N-1} a_n}_{s_k} + \sum_{n=N}^{N-1} (b_n - a_n) \end{aligned}$$

Taking limits, we conclude that

$$\lim_{k \rightarrow \infty} r_k = \lim_{k \rightarrow \infty} s_k + C$$

which concludes. ■

Proposition 1.25 (n^{th} term test)

If $\sum a_n$ converges, then $a_n \rightarrow 0$.

Remark 1.26. The converse does not hold. Consider the harmonic series. We have

$$s_k = \sum_{n=1}^k \frac{1}{n}$$

Observe

$$\begin{aligned} s_{2k} &= s_k + \frac{1}{n+1} + \cdots + \frac{1}{n+n} \\ &\geq s_k + \frac{1}{2n} + \cdots + \frac{1}{2n} \\ &= s_k + \frac{1}{2} \end{aligned}$$

Hence s_k is not Cauchy, so it does not converge.

Proof. $\sum a_n$ converges $\implies s_k$ converges, so (s_k) is Cauchy. Then $s_{k+1} - s_k = a_{k+1} \rightarrow 0$. ■

Let's first focus on tests for the convergence of $\sum a_n$, where $a_n \geq 0$ for all n .

Proposition 1.27 (Comparison test)

Suppose we have sequences satisfying $0 \leq b_n \leq a_n$ for all n . Then $\sum a_n$ converges $\implies \sum b_n$ converges.

Proof. Set s_k and r_k to be the partial sums of a_n, b_n respectively. Since $a_n, b_n \geq 0$, (s_k) and (r_k) are increasing. Furthermore, $s_k \rightarrow s \implies s_k \leq s$. But

$$b_n \leq a_n \implies \sum_{n=1}^k b_n \leq \sum_{n=1}^k a_n$$

so $r_k \leq s_k$, and hence r_k converges by the monotone convergence theorem. ■

Example 1.28

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. We have

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} = 1 + \sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$$

But then

$$0 \leq \frac{1}{(n+1)^2} \leq \frac{1}{n(n+1)}$$

and we know $\sum \frac{1}{n(n+1)} \rightarrow 1$. So $\frac{1}{n^2}$ converges by the comparison test.

Proposition 1.29 (Root test)

Suppose $a_n \geq 0$ for all n . Suppose there exists a such that $a = \lim_{n \rightarrow \infty} a_n^{1/n}$. If

- $a < 1$, then $\sum a_n$ converges;
- $a > 1$, then $\sum a_n$ diverges;
- $a = 1$, then the test is inconclusive.

Proof. If $a > 1$, then we have N such that $a_n^{1/n} > 1$ for any $n \geq N$. But then $a_n > 1$, so $a_n \not\rightarrow 0$, and we fail the n^{th} term test. Hence the series diverges.

If $a < 1$, then we can find $r \in \mathbb{R}$ with $a < r < 1$. Then we have N such that $a_n^{1/n} < r$ for any $n \geq N$. But then $a_n < r^n$ for all $n \geq N$. By the comparison test on the tails with a geometric series, $\sum a_n$ converges. ■

Example 1.30 • Consider $\sum \frac{1}{2^n}$, then $a_n^{1/n} = \frac{1}{2}$, so $\sum a_n$ converges.

- Consider $\sum 4^n$, then $a_n^{1/n} = 4$, so $\sum a_n$ does not converge.

Proposition 1.31 (Ratio test)

If $a_n \geq 0$ for all n and $\frac{a_{n+1}}{a_n} \rightarrow a$ for some finite a , then if

- $a < 1$, then $\sum a_n$ converges;
- $a > 1$, then $\sum a_n$ diverges;
- $a = 1$, then the test is inconclusive.

Example 1.32 • If we check $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$, we see both lie in the inconclusive regions for both tests.

- Let us consider $\sum \frac{n}{2^n}$. The ratio test gives

$$\frac{a_{n+1}}{a_n} = \frac{\frac{n+1}{2^{n+1}}}{\frac{n}{2^n}} = \frac{n+1}{2n} \rightarrow \frac{1}{2}$$

which is nice. If instead we use the root test, then

$$\left(\frac{n}{2^n}\right)^{\frac{1}{n}} = \frac{n^{1/n}}{2} \rightarrow \frac{1}{2}$$

But actually showing that $n^{1/n} \rightarrow 1$ is non-trivial... we need to apply l'Hôpital. Observe

$$n^{1/n} = e^{\log(n^{1/n})} = e^{\frac{1}{n} \log n} \rightarrow e^0 = 1$$

- If the ratio test is conclusive for a sequence, then the root test is also conclusive, but not the converse:

$$a_n = \begin{cases} 2^{-n} & n \text{ even} \\ 2^{-n+1} & n \text{ odd} \end{cases}$$

Then the limit of ratios does not exist, but the root test shows convergence.

Lecture n

Proposition 1.33 (Integral test)

Suppose $f : [1, \infty) \rightarrow [0, \infty)$ is a continuous decreasing function (and hence integrable in $[1, N]$ for each $N \in \mathbb{N}$), and define $a_n = f(n)$. Then

$$\sum a_n \text{ converges} \iff \lim_{n \rightarrow \infty} \int_1^n f(t) dt \text{ exists}$$

Furthermore, as $k \rightarrow \infty$,

$$\sum_{n=1}^k a_n - \int_1^k f(x) dx \rightarrow l \text{ for some } l \in [0, f(1)]$$

Proof. Observe (DIAGRAM)

$$\begin{aligned} s_k - a_1 &= \sum_{n=2}^k a_n \leq \int_1^k f(t) dt \\ s_{k-1} &= \sum_{n=1}^{k-1} a_n \geq \int_1^k f(t) dt \end{aligned}$$

Now

$\Rightarrow \int_1^k f(t) dt \leq s_{k-1} \leq s$ since s_k is increasing and converges by assumption. Hence $(\int_1^k f(t) dt)_k$ converges by MCT.

$\Leftarrow s_k \leq a_1 + \int_1^k f(t) dt \leq a_1 + \lim_{k \rightarrow \infty} \int_1^k f(t) dt$. Hence s_k converges by MCT.

For the last part, let

$$b_k = \sum_{n=1}^k a_n - \int_1^k f(x) dx$$

We have

$$b_k - b_{k-1} = a_k - \int_{k-1}^k f(t) dt = f(k) \cdot 1 - \int_{k-1}^k f(t) dt \leq 0$$

and

$$\begin{aligned} \sum_{n=1}^{k-1} a_n &\geq \int_1^k f(t) dt \\ \Rightarrow \sum_{n=1}^k a_n - \int_1^k f(t) dt &\geq a_k \\ \Rightarrow b_k &\geq a_k = f(k) \geq 0 \end{aligned}$$

Hence b_k is decreasing and bounded below, so converges to some $l \in \mathbb{R}$. Since $0 \leq b_k \leq b_1 = a_1 = f(1)$, we conclude $0 \leq l \leq f(1)$. $\blacksquare$

Example 1.34 • $\sum \frac{1}{n^p}$ converges if and only if $p > 1$, since

$$\lim_{x \rightarrow \infty} \int_1^x \frac{1}{t^p} dt = \lim_{x \rightarrow \infty} \begin{cases} \frac{1}{1-p} x^{1-p} & p \neq 1 \\ \log x & p = 1 \end{cases}$$

• $\sum \frac{1}{n \log n}$ diverges, since

$$\int \frac{1}{t \log t} dt = \log \log t + C$$

• $\sum \frac{1}{n \log^2 n}$ converges, since

$$\int \frac{1}{t \log^2 t} dt = -\frac{1}{\log t} + C$$

Proposition 1.35 (Cauchy condensation test, non-examinable)

Let $a_n \geq 0$ and (a_n) be decreasing. Then $\sum a_n$ converges if and only if $\sum 2^n a_{2^n}$ converges.

Proof. Observe

$$\begin{aligned} \sum_{m=1}^{2^n-1} a_m &= a_1 + a_2 + a_3 + \cdots + a_{2^n-1} \\ &\leq a_1 + 2a_2 + \cdots + 2^{n-1}a_{2^{n-1}} = \sum_{k=0}^{n-1} 2^k a_{2^k} \end{aligned}$$

and

$$\begin{aligned} \frac{1}{2} \sum_{k=0}^{n-1} 2^k a_{2^k} &= \frac{1}{2} a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{n-1} a_{2^{n-1}} \\ &\leq a_1 + a_2 + (a_3 + a_4) + (a_5 + \cdots + a_8) + \cdots + a_{2^{n-1}} = \sum_{m=1}^{2^n-1} a_m \end{aligned}$$

from which the result follows. ■

Proposition 1.36 (Alternating series test)

Let (a_n) be a decreasing sequence, and suppose $a_n \geq 0$, $a_n \rightarrow 0$. Then $\sum (-1)^{n+1} a_n$ converges.

Lecture $n + 1$

Proof. Set $s_k = \sum_{n=1}^k (-1)^{n+1} a_n$. Then

$$s_{2k} = \underbrace{(a_1 - a_2)}_{\geq 0} + \underbrace{(a_3 - a_4)}_{\geq 0} + \cdots + \underbrace{(a_{2k-1} - a_{2k})}_{\geq 0}$$

and so $s_2 \leq s_4 \leq \cdots$. Likewise, $s_1 \geq s_3 \geq \cdots$. Moreover, $s_{2k+1} - s_{2k} = a_{2k+1} \geq 0$, so that

$$s_2 \leq s_{2k} \leq s_{2k+1} \leq s_1$$

so s_{2k+1} is bounded below by s_2 , and s_{2k} is bounded above by s_1 . Hence, by the monotone convergence theorem, we have $s_{2k} \rightarrow s$, $s_{2k+1} \rightarrow \tilde{s}$. Now, note $s_{2k+1} - s_{2k} = a_{2k+1} \rightarrow 0$, and so $s = \tilde{s}$.

Finally, let us show s_n converges. We have for every $\varepsilon > 0$ an N_1 such that $|s_{2k} - s| < \varepsilon$ for any $k \geq N_1$, and an N_2 such that $|s_{2k+1} - s| < \varepsilon$ for any $k \geq N_2$. But then, taking $N = 2 \max(N_1, N_2) + 1$, we can see that $|s_k - s| < \varepsilon$ for any $k \geq N$. ■

Proposition 1.37 (Dirichlet test)

Let (a_n) be a decreasing sequence, and suppose $a_n \geq 0$, $a_n \rightarrow 0$. Let (b_n) be another sequence such that $s_k = (\sum_{n=1}^k b_n)$ is bounded. Then $\sum a_n b_n$ converges.

Proof. An exercise. ■

Definition 1.38 (Absolute convergence). We say $\sum a_n$ is *absolutely convergent* if $\sum |a_n|$ is convergent.

Lemma 1.39

If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

Proof. Let $s_k = \sum_{n=1}^k a_n$, $r_k = \sum_{n=1}^k |a_n|$. Observe

$$\begin{aligned} |s_k - s_l| &= \left| \sum_{n=l+1}^k a_n \right| \\ &\leq \sum_{n=l+1}^k |a_n| \\ &= r_k - r_l \\ &= |r_k - r_l| \end{aligned}$$

Hence r_k convergent $\implies r_k$ Cauchy $\implies s_k$ Cauchy $\implies s_k$ convergent. ■

Remark. This is awesome, because we can show absolute convergence using all our tests for series with positive terms (the converse is clearly false).

We sometimes call sequences that are convergent but not absolutely convergent conditionally convergent. Conditionally convergent sequences are weird.

Example 1.40

Consider

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

Rearranging terms on the right, we produce

$$\begin{aligned} &\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \left(\frac{1}{7} - \frac{1}{14}\right) + \dots \\ &= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \frac{1}{14} \\ &= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots\right) \end{aligned}$$

so we've produced a different value by rearranging the terms. For a conditionally convergent series, the order of the terms matters.

Proposition 1.41 (Rearrangements under absolute convergence)

Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Let $a'_n = a_{\sigma(n)}$. Then, if $\sum a_n$ is absolutely convergent, $\sum a'_n = \sum a_n$.

Proof. Let $s_k = \sum^k a_n$, and suppose $s_k \rightarrow s$. Then, for any ε we have N such that, for $k \geq N$,

$$|s_k - s| < \varepsilon \text{ and } \sum_{k > N} |a_n| < \varepsilon$$

Furthermore, there exists $N_1 \geq N$ such that $a_1, \dots, a_N$ are included in $a'_1, \dots, a'_{N_1}$. Hence, for $m \geq N_1$,

$$\begin{aligned} \sum_{n=1}^m a'_n &= \sum_{n=1}^N a_n + (m - N \text{ terms of } (a_n) \text{ with index } \geq N) \\ \Rightarrow \left| \sum_{n=1}^m a'_n - s \right| &\leq \underbrace{|s_N - s|}_{< \varepsilon} + \underbrace{\sum_{k > N} |a_n|}_{< \varepsilon} \\ &< 2\varepsilon \end{aligned}$$

which was what we wanted. ■

§2 Continuity of functions

§2.1 Limits of functions

Take $f : \mathbb{C} \rightarrow \mathbb{C}$. We want to define what we mean by $f(z) \rightarrow y$ as $z \rightarrow a$, even if $a \notin X$. A classic example is $\frac{\sin x}{x}$ at $a = 0$ – even though 0 is not in the domain, there are points in the domain that are very close to 0, so it feels like we should be able to assign 0 some value.

Definition 2.1 (Accumulation point). Let $X \subseteq \mathbb{C}$. We say a is an *accumulation point* for X if, for any $\delta > 0$, there exists $z \in X \setminus \{a\}$ such that $|z - a| < \delta$. Otherwise, we say a is *isolated*.

Example 2.2 • Take $X = \{0\} \cup [1, 2]$ – then 0 is an isolated point, but all points in $[1, 2]$ are accumulation points.

- Take $S = \{|z| = 1\}$ – these are accumulation points for the set $D = \{|z| < 1\}$ (also, every point in D is an accumulation point for D).

Lemma 2.3

Let $X \subseteq \mathbb{C}$, $a \in \mathbb{C}$. Then a is an accumulation point for X if and only if there exists (x_n) in $X \setminus \{a\}$ such that $x_n \rightarrow a$.

Let's return to thinking about $f(z) \rightarrow y$ as $z \rightarrow a$. We should be able to make sense of this if a is either in the domain of f , or is an accumulation point for the domain of f .

Definition 2.4. Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Let $a \in \mathbb{C}$ an accumulation point of X . Then we say $f(z) \rightarrow y$ as $z \rightarrow a$ if, for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon)$ such that $|z - a| < \delta, z \in X \setminus \{a\} \implies |f(z) - y| < \varepsilon$. We may also write

$$\lim_{z \rightarrow a} f(z) = y$$

Furthermore, if a is an accumulation point (but not in the domain), we say $f(z) \rightarrow \infty$ as $z \rightarrow a$ if

- $f(z)/|f(z)| \rightarrow l$ for some $l \in \mathbb{C}$;
- and, for any $L > 0$, there exists $\delta = \delta(L)$ such that $|z - a| < \delta, z \in X \setminus \{a\} \implies |f(z)| > L$.

Remark 2.5. If a is an isolated point of X , then we can always find δ sufficiently small such that $|z - a| < \delta, z \in X \iff z = a$. So we could define a limit for isolated points if we really wanted, but it wouldn't be very interesting.

Example 2.6

Take $f(x) = \frac{\sin x}{x}$ with domain $\mathbb{R} \setminus \{0\}$. We claim $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$.

Recall the geometric argument that shows $\cos x < \frac{\sin x}{x} < 1$ for every $x \in (0, \frac{\pi}{2})$. Hence

$$\left| \frac{\sin x}{x} - 1 \right| < |1 - \cos x| = 2 \sin^2 \left(\frac{x}{2} \right) < \frac{x^2}{2}$$

Hence, for any $\varepsilon > 0$, choosing $\delta = \sqrt{2\varepsilon}$ gives the result.

Lemma 2.7

Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$, and take $a \in \mathbb{C}$ an accumulation point of X . Then

- $f(z) \rightarrow y$ as $z \rightarrow a$ if and only if $f(z_n) \rightarrow y$ for every sequence (z_n) on $X \setminus \{a\}$ with $z_n \rightarrow a$;
- $f(z)$ diverges as $z \rightarrow a$ if and only if $f(z_n)$ diverges for every sequence (z_n) on $X \setminus \{a\}$ with $z_n \rightarrow a$.

Lemma 2.8

If $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ has a limit at $a \in \mathbb{C}$, then this limit is unique.

Proof. Suppose $f(z) \rightarrow x$ and $f(z) \rightarrow y$ as $z \rightarrow a$. Then

$$|x - y| \leq |x - f(z)| + |f(z) - y|$$

Taking the limit as $z \rightarrow a$, we have

$$|x - y| \leq 0 + 0$$

so $x = y$ as required. ■

Remark. This justifies the notation $\lim_{z \rightarrow a} f(z) = y$.

Lemma 2.9

Let $f, g : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$, and $a \in \mathbb{C}$ an accumulation point of X . Suppose $\lim_{z \rightarrow a} f(z) = y$, $\lim_{z \rightarrow a} g(z) = x$. Then

$$\lim_{z \rightarrow a} f(z) + g(z) = y + x$$

$$\lim_{z \rightarrow a} f(z)g(z) = yx$$

If $g(z) \neq 0$ for any $z \in X$, and $x \neq 0$, then

$$\lim_{z \rightarrow a} \frac{f(z)}{g(z)} = \frac{y}{x}$$

§2.2 Continuity of functions

Definition 2.10. Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is continuous at $a \in X$ if

$$f(a) = \lim_{z \rightarrow a} f(z)$$

or, equivalently, for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon)$ such that $|z - a| < \delta$, $z \in X \implies |f(z) - f(a)| < \varepsilon$.

If a is an isolated point, f is continuous at a , since $|z - a| < \delta \implies z = a$ for sufficiently small δ . If, on the other hand, a is an accumulation point, f is continuous at a if and only if

$$\lim_{z \rightarrow a} f(z) = f(a)$$

Definition 2.11. Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We say f is continuous if it is continuous at every point in X .

Example 2.12 • $f(z) = z$ is continuous – take $\delta = \varepsilon$.

- Consider

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

This function is not continuous. Take $x_n = \frac{1}{(2n + \frac{1}{2})\pi}$, which satisfies $x_n \rightarrow 0$ – but $f(x_n) = \sin\left((2n + \frac{1}{2})\pi\right) = 1$ for any n , so $f(x_n) \rightarrow 1 \neq f(0)$.

Lecture 8 – 10/02/26

There is another, equivalent notion of continuity which is extremely useful.

Definition 2.13 (Sequential continuity). Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. We say f is *sequentially continuous* at $a \in X$ if $f(z_n) \rightarrow f(a)$ for all (z_n) on X with $z_n \rightarrow a$.

Proposition 2.14

Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is continuous if and only if it is sequentially continuous.

Proof. $\implies$ By continuity of f , for any $\varepsilon > 0$ we have δ such that $|z - a| < \delta$, $z \in X$ implies $|f(z) - f(a)| < \varepsilon$. Take (z_n) on X with $z_n \rightarrow a$, then there exists $N = N(\delta(\varepsilon))$ such that, for any $n \geq N$, $|z_n - a| < \delta$ and hence $|f(z_n) - f(a)| < \varepsilon$.

$\impliedby$ Let us take the contrapositive. If f is not continuous at a , then there exists $\varepsilon > 0$ suchb that, for any $\delta > 0$, one can find $z = z(\delta) \in X$ such that $|f(z) - f(a)| \geq \varepsilon$ while $|z - a| < \delta$. Now take $\delta = 1, \frac{1}{2}, \dots, \frac{1}{n}, \dots$ to find (z_n) with $|z_n - a| < \frac{1}{n}$, so $z_n \rightarrow a$, but $|f(z_n) - f(a)| \geq \varepsilon$ for all n and a non-zero ε . Hence f cannot be sequentially continuous, as required. ■

Example

Let us have some more examples.

- Consider the Dirichlet function $f(x) = 1_{\mathbb{Q}}(x)$. This is discontinuous at every real a :
 - if $a \in \mathbb{Q}$, we can find $(x_n) \subset \mathbb{R} \setminus \mathbb{Q}$ with $x_n \rightarrow a$, but of course $f(x_n) = 0$, so $f(x_n) \rightarrow 0 \neq 1 = f(a)$;
 - if $a \in \mathbb{R} \setminus \mathbb{Q}$, we proceed analogously.
- $f(x) = \sin x$ is continuous as a real function. Fix $a \in \mathbb{R}$ and, for some $\varepsilon > 0$, pick $\delta = \min(\varepsilon, \pi)$. Now

$$\begin{aligned}
 |f(x) - f(a)| &= |\sin x - \sin a| \\
 &= 2 \left| \cos\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right) \right| \\
 &\leq 2 \left| \sin\left(\frac{x-a}{2}\right) \right| \\
 &\leq |x - a| < \varepsilon
 \end{aligned}$$

since $\frac{x-a}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Now let us look at some important properties of continuous functions.

Lemma 2.15

Let $f, g : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ be continuous at $a \in \mathbb{C}$. Then, so are $\lambda f + g$ for $\lambda \in \mathbb{C}$ fixed, fg and, if $f(z) \neq 0$ for any $z \in X$, $1/f(z)$.

Lemma 2.16

Let $U, V \subseteq \mathbb{C}$, and $f : U \rightarrow V$, $g : V \rightarrow \mathbb{C}$. Suppose f is continuous at $a \in U$ and g is continuous at $f(a) \in V$ – then, $g \circ f : U \rightarrow \mathbb{C}$ is continuous at a .

Proof. Since g is continuous at $f(a)$, we have $\sigma = \sigma(\varepsilon)$ such that $y \in V$, $|y - f(a)| < \sigma$ implies $|g(y) - g(f(a))| < \varepsilon$. Furthermore, since f is continuous at a , we have for any σ a $\delta = \delta(\sigma) = \delta(\varepsilon)$ such that

$$z \in V, |z - a| < \delta \implies |f(z) - f(a)| < \sigma \implies |g(f(z)) - g(f(a))| < \varepsilon$$

■

§2.3 Extreme value theorem

Definition 2.17 (Closed sets). A set $X \subseteq \mathbb{C}$ is *closed* if all sequences (x_n) in X which converge in $\mathbb{C}$ have limits in X .

- Example**
- The closed interval $[0, 1]$ is closed, while the open interval $(0, 1)$ is not.
 - $[0, 1)$ and $(0, 1]$ are not closed.
 - $[0, \infty)$ is closed, while $(0, \infty]$ is not.

Definition 2.18 (Bounded sets). We say $X \subseteq \mathbb{C}$ is *bounded* if there exists $M > 0$ such that $X \subseteq \{z \in \mathbb{C} : |z| \leq M\}$, or in other words $\sup_{z \in X} |z| \leq M$.

- Example**
- $[0, 1]$, $(0, 1)$, $[0, 1)$ and $(0, 1]$ are all bounded, while $[0, \infty)$ and $(0, \infty)$ are not.

Proposition 2.19

Let $X \subseteq \mathbb{C}$ be a closed bounded set. If $f : X \rightarrow \mathbb{C}$ is continuous, then $f(X) \subseteq \mathbb{C}$ is a closed bounded subset of $\mathbb{C}$.

Proof.

- $f(X)$ is bounded. Suppose not. Then, for each $n \in \mathbb{N}$, we can find $x_n \in X$ such that $|f(x_n)| > n$. Now, (x_n) is a sequence in X , so it must be bounded, so Bolzano-Weierstrass applies – there exists a convergent subsequence (x_{n_k}) such that $x_{n_k} \rightarrow x$ and, by closedness of X , $x \in X$. On the other hand, $|f(x_{n_k})| > n_k \geq k$, so $f(x_{n_k})$ cannot converge as $k \rightarrow \infty$. But then f is not continuous, since it is not sequentially continuous.

- $f(X)$ is closed. Take (y_n) in $f(X)$, and suppose it converges to some $y \in \mathbb{C}$ – we want to show $y \in f(X)$. Since $y_n \in f(X)$, we have $x_n \in X$ such that $f(x_n) = y_n$, and (x_n) has a subsequence (x_{n_k}) converging to $x \in X$ by the closure of X and the Bolzano-Weierstrass theorem. By continuity of f ,

$$y = \lim_{k \rightarrow \infty} y_{n_k} = \lim_{k \rightarrow \infty} f(x_{n_k}) = f\left(\lim_{k \rightarrow \infty} x_{n_k}\right) = f(x)$$

which was what we wanted.

■

Theorem 2.20 (Extreme value theorem)

Let $X \subseteq \mathbb{R}$ be a closed bounded set, and let $f : X \rightarrow \mathbb{R}$ be continuous. Then there exist $a, b \in X$ satisfying

$$f(a) = \sup f(X); \quad f(b) = \inf f(X)$$

Lecture 9 – 12/02/26

Proof. We will consider specifically the supremum, since the infimum is analogous.

Let $M = \sup f(X)$, which exists since $f(X)$ is bounded by [proposition 2.19](#). By definition, $M - \frac{1}{n}$ is not an upper bound on $f(X)$ for any $n \in \mathbb{N}$, that is, we can find a sequence (y_n) with elements in $f(X)$ such that $M - \frac{1}{n} < y_n \leq M$, which produces a sequence (x_n) on X such that $M - \frac{1}{n} < f(x_n) \leq M$. Now take limits and use the fact that f is continuous – we have $M \leq \lim_{n \rightarrow \infty} f(x_n) \leq M$.

Finally, by [proposition 2.19](#), we know that, $f(X)$ is closed, so, since (y_n) is a sequence in $f(X)$, its limit is also in the image of X . ■

Remark. We know that the sup and inf exist – the key point is that f actually achieves both of these.

§2.4 Intermediate value theorem**Theorem 2.21 (Intermediate value theorem)**

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous then, for any y satisfying $f(a) \leq y \leq f(b)$, there exists x satisfying $a \leq x \leq b$ such that $f(x) = y$.

Proof. If f is constant, $y = f(a)$ or $y = f(b)$, then the result is clear. If not, wlog let $f(a) < y < f(b)$. Set $S = \{x \in [a, b] : f(x) \leq y\}$. Then $a \in S$, so S is non-empty – furthermore, S is bounded above by b , so we have $c = \sup S \in [a, b]$. We claim that $f(c) = y$. Observe:

- We claim $f(c) \leq y$. Suppose not. Then $\varepsilon = f(c) - y > 0$. By continuity of f , we have δ such that $|x - c| < \delta$, $x \in [a, b]$ implies $|f(x) - f(c)| < \varepsilon$, so $f(x) > f(c) - \varepsilon = y$, so $x \notin S$. This means $(c - \delta, c) \cap S = \emptyset$, which contradicts the definition of c (in particular that c is the least upper bound).
- We claim $f(c) \geq y$. Suppose not. Then $\varepsilon = y - f(c) > 0$. By continuity of f , we have δ such that $|x - c| < \delta$, $x \in [a, b]$ implies $|f(x) - f(c)| < \varepsilon$, so $f(x) < f(c) + \varepsilon = y$. But then $f(c + \frac{\delta}{2}) < y$, so $c \leq c + \frac{\delta}{2} \in S$, which contradicts the definition of c (in particular that c is an upper bound).

The result follows. ■

Remark. A nice interpretation of this theorem is that continuous functions from reals to reals map intervals to intervals.

Example

We can use this result to show the existence of n^{th} roots. Take $t > 0$ and $N \in \mathbb{N}$, and consider

$$\begin{aligned} f : [0, 1+t] &\rightarrow \mathbb{R} \\ x &\mapsto x^N \end{aligned}$$

This is continuous, and $f(0) = 0 < t < (1+t)^N = f(1+t)$. Hence, there exists a c satisfying $c^N = t$ by the intermediate value theorem. Note that we do not have uniqueness.

Definition 2.22 (Monotone functions). A function $f : [a, b] \rightarrow \mathbb{R}$ is *monotone* if it is either increasing, $x_1 \leq x_2 \implies f(x_1) \leq f(x_2)$; or decreasing, $x_1 \leq x_2 \implies f(x_2) \leq f(x_1)$.

Proposition 2.23 (Inverse function theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and strictly monotone. Let $c = \min(f(a), f(b))$, $d = \max(f(a), f(b))$. Then $f : [a, b] \rightarrow [c, d]$ is a bijection, and $f^{-1} : [c, d] \rightarrow [a, b]$ is continuous and strictly monotone.

Proof. Since f is continuous, it achieves its extrema by [theorem 2.20](#); since it is monotone, it achieves them at its endpoints, so it maps $[a, b] \rightarrow [c, d]$ – since it is strictly monotone, it does so injectively. By [theorem 2.21](#), it is surjective, and hence it is bijective.

For the rest of the proof, we suppose wlog that f is strictly increasing. Suppose for contradiction that $f^{-1} : [c, d] \rightarrow [a, b]$ is not strictly increasing – then we have $y_1, y_2 \in [c, d]$ with $y_1 < y_2$ and $f^{-1}(y_1) \geq f^{-1}(y_2)$. But then, applying f to both sides, we obtain $y_1 = f(f^{-1}(y_1)) \geq f(f^{-1}(y_2)) = y_2$, which is a contradiction.

Finally, it remains to show f^{-1} is continuous. Let $y_0 \in [c, d]$ and $x_0 = f^{-1}(y_0)$. We split into three cases.

- $x_0 \in (a, b)$. Fix $\varepsilon > 0$, and choose $\eta \in (0, \varepsilon]$ sufficiently small for $[x_0 - \eta, x_0 + \eta] \subseteq [a, b]$. Now, since f is strictly increasing,

$$\begin{aligned} x_0 - \eta &< x_0 < x_0 + \eta \\ \implies f(x_0 - \eta) &< f(x_0) = y_0 < f(x_0 + \eta) \end{aligned}$$

Let us now take $\delta = \min(f(x_0 + \eta) - y_0, y_0 - f(x_0 - \eta)) > 0$. Suppose $|y - y_0| < \delta$, then

$$\begin{aligned} f(x_0 - \eta) &\leq y_0 - \delta < y < y_0 + \delta \leq f(x_0 + \eta) \\ \implies f(x_0 - \eta) &< y < f(x_0 + \eta) \\ \implies |f^{-1}(y) - x_0| &< \eta \\ \implies |f^{-1}(y) - f^{-1}(y_0)| &< \varepsilon \end{aligned}$$

as required.

- $x_0 = a, y_0 = f(a)$. Fix $\varepsilon > 0$, and choose $\eta = \min(\varepsilon, b-a)$. Set $\delta = f(a+\eta) - f(a) > 0$. Then $|y - y_0| < \delta, y \in [c, d]$ gives

$$\begin{aligned} f(a) &\leq y < f(a+\eta) \\ \implies a &\leq f^{-1}(y) < a+\eta \\ \implies |f^{-1}(y) - f^{-1}(y_0)| &< \eta \leq \varepsilon \end{aligned}$$

as required.

- $x_0 = b$ proceeds analogously.

This completes the result. ■

Lecture 10 – 14/02/26

§3 Differentiation

§3.1 Basics

Definition 3.1 (Derivative). Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}, a \in X$. We say f is differentiable at a if the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. We call this limit the derivative of f at a , $f'(a)$ or $\frac{df}{dx}(a)$.

Remark. We can't really make sense of the derivative at an isolated point of the domain. For accumulation points, we can distinguish

- interior points: we can approach from every direction and we need the limit to be direction-independent;
- non-interior points: the limit is 'domain-restricted'.

Example • $f(z) = z$ is differentiable everywhere:

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = \lim_{h \rightarrow 0} 1 = 1$$

• $f(z) = \bar{z}$ is not differentiable anywhere. Indeed,

$$\begin{aligned} h = \lambda \in \mathbb{R} : \lim_{\lambda \rightarrow 0} \frac{f(z+\lambda) - f(z)}{\lambda} &= \lim_{\lambda \rightarrow 0} \frac{\overline{z+\lambda} - \bar{z}}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{\lambda}{\lambda} = 1 \end{aligned}$$

$$\begin{aligned} h = i\lambda, \lambda \in \mathbb{R} : \lim_{\lambda \rightarrow 0} \frac{f(z+i\lambda) - f(z)}{i\lambda} &= \lim_{\lambda \rightarrow 0} \frac{\overline{z+i\lambda} - \bar{z}}{i\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{-i\lambda}{i\lambda} \\ &= -1 \end{aligned}$$

Hence, the limit is different depending on the direction we approach from, and so does not exist.

• $f(z) = \sin z$ is differentiable. Observe

$$\begin{aligned} f'(x) &\stackrel{?}{=} \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin h \cos x}{h} + \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1)}{h} \\ &= \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} + \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} \end{aligned}$$

The limit of $\frac{\sin h}{h}$ is 1 as already established, and the limit of $\frac{\cos h - 1}{h}$ can be shown to be zero by, for example, using a double angle formula.

Lemma 3.2

Let $f, g : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ be differentiable at $a \in X$. Then so are $f + g$, fg and, if $f(z) \neq 0$ for any $z \in X$, so is $1/f$. Furthermore, we have

$$\begin{aligned} (f+g)'(a) &= f'(a) + g'(a) \\ (fg)'(a) &= f'(a)g(a) + f(a)g'(a) \\ \left(\frac{1}{f}\right)'(a) &= -\frac{f'(a)}{f(a)^2} \end{aligned}$$

Proof. The addition rule is an exercise. For the multiplication rule, we have

$$\begin{aligned} (fg)'(a) &= \lim_{h \rightarrow 0} \frac{(fg)(a+h) - (fg)(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \end{aligned}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{(f(a+h) - f(a))g(a+h) + (f(a+h) - g(a))f(a)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(f(a+h) - f(a))g(a+h)}{h} + f(a) \lim_{h \rightarrow 0} \frac{g(a+h) - g(a)}{h} \\
&= \lim_{h \rightarrow 0} g(a+h)f'(a) + g'(a)f(a)
\end{aligned}$$

We will later show that differentiability implies continuity, which will allow us to conclude. ■

Example

Let us consider some examples.

- $f(z) = z^2 = z \cdot z$ is differentiable with $f'(z) = 2z$. By induction, z^n is differentiable with $f'(z) = nz^{n-1}$, so polynomials are differentiable.
- $f(z) = \frac{1}{z}$ is differentiable on $\mathbb{C} \setminus \{0\}$ with $f'(z) = -\frac{1}{z^2}$. By induction, $\frac{1}{z^n}$ has derivative $-\frac{n}{z^{n+1}}$.

More generally, rational functions $\frac{p(x)}{q(x)}$ with p, q polynomials are differentiable wherever $q(x) \neq 0$.

Lecture 11 – 17/02/26

Proposition 3.3 (Chain rule)

Let $U, V \subseteq \mathbb{C}$ and $f : U \rightarrow V, g : V \rightarrow \mathbb{C}$. Suppose that f is differentiable at $a \in U$ and g is differentiable at $f(a) \in V$. Then $g \circ f : U \rightarrow \mathbb{C}$ is differentiable at $a \in U$, and $(g \circ f)'(a) = g'(f(a))f'(a)$.

We will first need a lemma.

Lemma 3.4

Let $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is differentiable at $a \in X$ if and only if there exists $A \in \mathbb{C}$ and $\xi : \{z : z + a \in X\} \rightarrow \mathbb{C}$ satisfying $\xi(h) \rightarrow 0$ as $h \rightarrow 0$ such that

$$f(a+h) = f(a) + Ah + \xi(h)|h|$$

Furthermore, $A = f'(a)$.

Remark. In other words, $f(a+h) \approx f(a) + Ah$, and the function $\xi(h)$ allows us to quantify that error. In little o notation, this statement is $f(a+h) = f(a) + Ah + o(h)$.

Proof of lemma. $\Leftarrow$ We have

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{Ah + \xi(h)|h|}{h}$$

$$\begin{aligned}
&= A + \lim_{h \rightarrow 0} \left(\xi(h) \frac{|h|}{h} \right) \\
&= A
\end{aligned}$$

since $\frac{|h|}{h}$ is bounded and $\xi(h) \rightarrow 0$ as $h \rightarrow 0$. Hence the limit exists and f is differentiable as required.

$\implies$ Let $A = f'(a)$, so

$$\begin{aligned}
&\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = A \\
\implies \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - Ah}{h} &= 0
\end{aligned}$$

Now take

$$\xi(h) = \begin{cases} \frac{f(a+h) - f(a) - Ah}{|h|} & h \neq 0 \\ 0 & h = 0 \end{cases}$$

which satisfies $\xi(h) \rightarrow 0$ as $h \rightarrow 0$. ■

Proof of proposition. Since f, g are differentiable at $a, f(a)$ respectively, there exist error functions ξ_f, ξ_g satisfying $\xi_f(h) \rightarrow 0$ as $h \rightarrow 0$ and $\xi_g(k) \rightarrow 0$ as $k \rightarrow 0$, and

$$\begin{aligned}
f(a+h) &= f(a) + f'(a)h + \xi_f(h)|h| \\
g(f(a)+k) &= g(f(a)) + g'(f(a))k + \xi_g(k)|k|
\end{aligned}$$

Now

$$\begin{aligned}
(g \circ f)(a+h) - (g \circ f)(a) &= g(f(a+h)) - g(f(a)) \\
&= g(f(a) + \underbrace{f'(a)h + \xi_f(h)|h|}_k) - g(f(a)) \\
&= g(f(a)) + g'(f(a)) (f'(a)h + \xi_f(h)|h|) + \xi_g(k)|k| - g(f(a)) \\
&= g'(f(a))f'(a)h + g'(f(a))\xi_f(h)|h| + \xi_g(k)|k|
\end{aligned}$$

so we want to make

$$\begin{aligned}
\xi(h) &= g'(f(a))\xi_f(h) + \xi_g(k) \frac{|k|}{|h|} \\
&= g'(f(a))\xi_f(h) + \xi_g(hf'(a) + \xi_f(h)|h|) \cdot |f'(a) + \xi_f(h)|
\end{aligned}$$

It remains only to check that $\xi(h) \rightarrow 0$ as $h \rightarrow 0$. We have

$$\begin{aligned}
\lim_{h \rightarrow 0} \xi(h) &= \lim_{h \rightarrow 0} (g'(f(a))\xi_f(h) + \xi_g(hf'(a) + \xi_f(h)|h|) \cdot |f'(a) + \xi_f(h)|) \\
&= \lim_{h \rightarrow 0} g'(f(a))\xi_f(h) + \lim_{h \rightarrow 0} \xi_g(hf'(a) + \xi_f(h)|h|) \cdot \lim_{h \rightarrow 0} |f'(a) + \xi_f(h)|
\end{aligned}$$

Then the first term is zero because $\xi_f(h) \rightarrow 0$ as $h \rightarrow 0$, and the first multiplicand in the second term goes to 0 as $h \rightarrow 0$ because it is clear that its argument goes to 0 as $h \rightarrow 0$. ■

Example

Take

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

At $x \neq 0$, the chain and product rules give

$$\begin{aligned} f'(x) &= 1 \cdot \sin\left(\frac{1}{x}\right) + x \cdot \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) \\ &= \sin\left(\frac{1}{x}\right) - \frac{1}{x} \cos\left(\frac{1}{x}\right) \end{aligned}$$

At $x = 0$, the definition gives

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h} = \lim_{h \rightarrow 0} \sin\left(\frac{1}{h}\right)$$

which does not exist – hence the function is not differentiable at 0.

Lemma 3.5

If $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is differentiable at $a \in X$, then it is continuous at $a \in X$.

Proof. By lemma 3.4, we have

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{h \rightarrow 0} f(a + h) = \lim_{h \rightarrow 0} f(a) + Ah + \xi(h)|h| \\ &= f(a) + A \lim_{h \rightarrow 0} h + \lim_{h \rightarrow 0} (\xi(h)|h|) \\ &= f(a) \end{aligned}$$

■

Lecture 12 – 19/02/26**§3.2 Mean value theorems**

We think of the derivative as an instantaneous rate of change, and we want to try and relate this to *average* rates of change. Let us first have a proposition-lemma.

Proposition 3.6 (Rolle's theorem)

Let f cts: $[a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) , and suppose $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. By theorem 2.20, f attains its min and max, that is there exist x_m and $x_M \in [a, b]$

with

$$\sup_{[a,b]} f = f(x_M) \quad \inf_{[a,b]} f = f(x_m)$$

If f is constant, then $f'(c) = 0$ for any $c \in (a, b)$, so we are done. Otherwise, it either holds that $f(x_M) > f(a)$ (so $x_M \in (a, b)$) or $f(x_m) < f(a)$ (so $x_m \in (a, b)$). It remains to show that, at either of these points, $f' = 0$. We take cases.

- $x_m \in (a, b)$. Then, for sufficiently small h ,

$$\text{sign} \left(\frac{f(x_m + h) - f(x_m)}{h} \right) = \text{sign}(h)$$

so, taking $h \rightarrow 0$ from the right and left, we conclude $f'(x_m)$ is both positive and negative respectively, i.e. $f'(x_m) = 0$.

- $x_M \in (a, b)$. Then we have

$$\text{sign} \left(\frac{f(x_M + h) - f(x_M)}{h} \right) = -\text{sign}(h)$$

which again shows $f'(x_M) = 0$ by taking limits on both sides.

■

Theorem 3.7 (Mean value theorem)

Let f cts: $[a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) . Then, there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. Let

$$l(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

in other words, the secant line connecting $(a, f(a))$ to $(b, f(b))$. Then let $\phi(x) = f(x) - l(x)$, so $\phi(a) = \phi(b) = 0$. Hence Rolle gives c satisfying $\phi'(c) = 0$, that is

$$\begin{aligned} f'(c) - \frac{f(b) - f(a)}{b - a} &= 0 \\ \implies f'(c) &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

as required.

■

Corollary 3.8

Let f cts: $[a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) . Given h such that $a + h \in [a, b]$, there exists $\theta \in (0, 1)$ such that

$$f(a + h) = f(a) + hf'(a + \theta h)$$

Proof. Restrict the domain of f to $[a, a+h]$. Then, by the mean value theorem, we have $c \in [a, a+h]$ with

$$f'(c) = \frac{f(a+h) - f(a)}{h} \\ \implies f(a+h) = f(a) + hf'(c)$$

Then, let θ be such that $c = a + \theta h$. ■

Corollary 3.9

Let f cts : $[a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) . Then

- $f' \geq 0$ on $(a, b) \implies f$ is increasing, resp. strictly increasing;
- $f' \leq 0$ on $(a, b) \implies f$ is decreasing, resp. strictly decreasing;
- $f' = 0$ on $(a, b) \implies f$ is constant.

Proof. Exercise. ■

Remark. Statements such as [corollary 3.9](#) do not necessarily hold if the domain X is an arbitrary subset of $\mathbb{R}$. For instance, take

$$f : \mathbb{Q} \rightarrow \mathbb{R} \\ x \mapsto \begin{cases} 0 & x^2 > 2 \\ 1 & x^2 < 2 \end{cases}$$

which is continuous (yes, really!) and satisfies $f'(x) = 0$ for any $x \in \mathbb{Q}$, but is not constant.

Lemma 3.10

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be differentiable ($\implies$ continuous) in $\mathbb{C}$. Then, $f'(z) = 0$ for all $z \in \mathbb{C} \implies f$ constant.

Proof. We want to reduce to the $\mathbb{R}$ case. For some $z \in \mathbb{C}$, take $g(t) = f(tz)$ where $g : [0, 1] \rightarrow \mathbb{C}$. Note g is continuous on $[0, 1]$, g is differentiable on $(0, 1)$, so we can apply [theorem 3.7](#) and its corollaries to $\operatorname{Re}(g(t))$ and $\operatorname{Im}(g(t))$:

$$g'(t) = zf'(tz) = 0 \implies \begin{cases} \operatorname{Re}(g'(t)) = 0 \\ \operatorname{Im}(g'(t)) = 0 \end{cases}$$

and so $g(t)$ is constant – hence $f(0) = g(0) = g(1) = f(z)$. But z was arbitrary, so $f(z) = f(0)$ for any $z \in \mathbb{C}$. ■

Remark. We can use similar ideas to show a version of the mean value theorem in $\mathbb{C}$.

Theorem 3.11 (Inverse function theorem – differentiable version)

Let f cts: $[a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) . Assume that $f'(x) > 0$ for any $x \in (a, b)$. Then $f : [a, b] \rightarrow [f(a), f(b)]$ is a bijection with f^{-1} cts: $[f(a), f(b)] \rightarrow [a, b]$ differentiable on $(f(a), f(b))$ with

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$$

Proof. By [corollary 3.9](#), f is strictly increasing, so our previous version of IFT, [proposition 2.23](#), gives that f is a bijection and has a continuous inverse f^{-1} . It remains to show that f^{-1} is differentiable and satisfies the above.

Let $y \in (f(a), f(b))$ and $x = f^{-1}(y)$. Given h such that $y + h \in (f(a), f(b))$, define k such that $y + h = f(x + k)$, or equivalently $k = f^{-1}(y + h) - x$. Then

$$\begin{aligned} \frac{f^{-1}(y + h) - f^{-1}(y)}{h} &= \frac{x + k - x}{f(x + k) - y} \\ &= \frac{k}{f(x + k) - f(x)} \\ &= \frac{1}{\frac{f(x + k) - f(x)}{k}} \end{aligned}$$

so the result holds so long as $k \rightarrow 0$ as $h \rightarrow 0$ – then we can take limits on both sides. But

$$\begin{aligned} \lim_{h \rightarrow 0} k &= \lim_{h \rightarrow 0} f^{-1}(y + h) - f^{-1}(y) \\ &= 0 \end{aligned}$$

because f^{-1} is continuous. ■

Example 3.12

Let us consider some examples.

- Fix $R > 0$, and define

$$\begin{aligned} f : [0, R] &\rightarrow \mathbb{R} \\ x &\rightarrow x^N \end{aligned}$$

so f is continuous on $[0, R]$ and differentiable on $(0, R)$ with $f'(x) = Nx^{N-1}$. Hence, by [theorem 3.11](#), $f : [0, R] \rightarrow [0, R^N]$ is a bijection, and $f^{-1}(y) = y^{1/N}$ is differentiable with

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} = \frac{1}{N(f^{-1}(y))^{N-1}} = \frac{1}{N}y^{\frac{1}{N}-1}$$

- $f(x) = e^x$ is continuous and differentiable on $\mathbb{R}$ with $f'(x) = e^x > 0$. Hence, we have an inverse $f^{-1}(y) = \log y$ such that

$$(f^{-1})'(y) = \frac{1}{e^{f^{-1}(y)}} = \frac{1}{e^{\log y}} = \frac{1}{y}$$

Proposition 3.13 (Cauchy's mean value theorem)

Let f, g cts: $[a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))$$

Proof. We will reduce to Rolle's theorem, [proposition 3.6](#). Let

$$\begin{aligned} \phi(x) &= (g(x) - g(a))(f(b) - f(a)) - (g(b) - g(a))(f(x) - f(a)) \\ \implies \phi'(x) &= g'(x)(f(b) - f(a)) - f'(x)(g(b) - g(a)) \end{aligned}$$

and $\phi(a) = \phi(b) = 0$. The result follows. ■

Proposition 3.14 (L'Hôpital's rule, informal statement)

For certain limits expressed as a ratio of functions f and g , one can calculate the limit by taking derivatives first.

Lecture 13 – 21/02/26**§3.3 Higher derivatives and Taylor's theorem**

Definition 3.15 (Higher derivatives). Suppose $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is differentiable on X . We say f is n -times differentiable on X if f is $(n - 1)$ -times differentiable on X and

$$\begin{aligned} f^{(n-1)} : X &\rightarrow \mathbb{C} \\ x &\mapsto f^{(n-1)}(x) \end{aligned}$$

is differentiable.

Definition 3.16 (Continuous differentiability). We say $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is k -times continuously differentiable on X , and write $f \in C^k(X)$, if f is k -times differentiable on X and $f^{(k)}$ is continuous.

Remark. Recall that f differentiable $\implies f$ continuous, so f k -times differentiable $\implies f$ $(k - 1)$ -times continuously differentiable.

Definition 3.17 (Smoothness). We say $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is *smooth* if it is k -times differentiable for any k , or, equivalently, if $f \in C^k(X)$ for any k .

Since the derivative of a function at a point x_0 determines the slope of the tangent, $f(x_0)$ and $f'(x_0)$ together yield a linear approximation to the graph of f at x_0 :

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

for $x \approx x_0$. If we have more derivatives, however, we should in principle be able to get a better approximation:

$$\begin{aligned} f'(x) &\approx f'(x_0) + f''(x_0)(x - x_0) \\ \implies f(x) &\approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 \end{aligned}$$

and, if f is n -times differentiable,

$$f(x) \approx \sum_{k=0}^n \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k$$

for $x \approx x_0$.

Definition 3.18 (Taylor polynomials and Taylor series). If $f : X \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is n -times differentiable and $x_0 \in X$, then

$$T_{n,f,x_0}(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k$$

is called the n^{th} -order Taylor polynomial of f at x_0 . If f is smooth, then

$$T_{f,x_0}(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k$$

is called the Taylor series of f at x_0 .

We would like to quantify exactly how good this approximation is.

Theorem 3.19 (Taylor's theorem: Lagrange remainder)

Suppose $f : X \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and its $n - 1$ derivatives are continuous on $[a, a + h]$ and f is n -times differentiable on $(a, a + h)$. Set

$$R_{n,f,a}(h) = f(a + h) - \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(a)h^k = f(a + h) - T_{n-1,f,a}(h)$$

Then, there exists $\theta(h) \in (0, 1)$ such that

$$R_{n,f,a}(h) = \frac{1}{n!} h^n f^{(n)}(a + \theta h)$$

Remark. The conditions for the above are a consequence of $f \in C^n((c, d))$ for any $(c, d) \supset [a, a + h]$.

Morally speaking, we are saying that $R_{n,f,a}(h) = O(h^n)$ as $h \rightarrow 0$, and we know exactly what the constant is. Unfortunately, this does *not* tell us anything about whether $R_{n,f,a}(h) \rightarrow 0$ as $n \rightarrow \infty$: even if $f \in C^\infty(X)$, we can't say how the scaling factor of $M_n = \sup_{x \in [a, a+h]} |f^{(n)}(x)|$ behaves with n . THIS PART NEEDS TO BE UPDATED!!

Remark. We should also note that there is nothing important about $h > 0$ – if $h = -t < -0$, and f and its $n - 1$ derivatives are continuous and $[a - t, a]$ and f is n -times differentiable on $(a - t, a)$, then we can apply the above to $g(x) = f(-x)$ to obtain

$$\begin{aligned} g(-a + t) &= \sum_{k=0}^{n-1} \frac{(-1)^k}{k!} f^{(k)}(a) t^k + \frac{(-1)^n}{n!} t^n f^{(n)}(a - \theta t) \\ \implies f(a - t) &= \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(a) (-t)^k + \frac{1}{n!} (-t)^n f^{(n)}(a + \theta(-t)) \end{aligned}$$

Proof. Without loss of generality, we may take $a = 0$ (otherwise apply to $g(x) = f(x + a)$). Let

$$\phi : [0, h] \rightarrow \mathbb{R}$$

$$t \mapsto f(t) - \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(0) t^k - \frac{t^n}{n!} B = f(t) - T_{n-1, f, 0}(t) - \frac{t^n}{n!} B$$

where B is chosen such that $\phi(h) = 0$. Note ϕ and its first $n - 1$ derivatives are continuous on $[0, h]$, and its n^{th} derivative exists on $(0, h)$. One can see fairly easily that $\phi^{(k)}(0) = 0$ for $k = 0, \dots, n - 1$. We can thus apply [corollary 3.8](#) repeatedly:

$$\begin{aligned} \phi(0) = \phi(h) = 0 &\implies \exists \theta_1 \in (0, 1) \text{ s.t. } \phi'(\theta_1 h) = 0 \\ \phi'(0) = \phi'(\theta_1 h) = 0 &\implies \exists \theta_2 \in (0, 1) \text{ s.t. } \phi''(\theta_2 \theta_1 h) = 0 \\ &\vdots \\ \phi^{(n-1)}(0) = \phi^{(n-1)}(\theta_{n-1} \dots \theta_1 h) = 0 &\implies \exists \theta_n \in (0, 1) \text{ s.t. } \phi^{(n)}(\theta_n \dots \theta_1 h) = 0 \end{aligned}$$

Now, setting $\theta = \theta_n \dots \theta_1 \in (0, 1)$, we end up with $\theta \in (0, 1)$ such that $\phi^{(n)}(\theta h) = f^{(n)}(\theta h) - B = 0$, so $B = f^{(n)}(\theta h)$ as required. $\blacksquare$

Proof (alternative). Again wlog let us take $a = 0$. Now let

$$g : [0, h] \rightarrow \mathbb{R}$$

$$t \mapsto f(h) - \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k)}(t) (h - t)^k = f(h) - T_{n-1, f, 0}(h - t)$$

Note that g is continuous on $[0, h]$ and differentiable on $(0, h)$ with

$$\begin{aligned} g'(t) &= - \sum_{k=1}^{n-1} \frac{1}{(k-1)!} f^{(k)}(t) (-1) (h - t)^{k-1} - \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k+1)}(t) (h - t)^k \\ &= - \sum_{k=0}^{n-2} \frac{1}{k!} f^{(k+1)}(t) (h - t)^k - \sum_{k=0}^{n-1} \frac{1}{k!} f^{(k+1)}(t) (h - t)^k \\ &= - \frac{1}{(n-1)!} f^{(n)}(t) (h - t)^{n-1} \end{aligned}$$

For $p \in \{1, \dots, n\}$, set

$$\phi_p(t) = g(t) - \left(\frac{h-t}{h} \right)^p g(0)$$

Then $\phi_p(0) = \phi_p(h) = 0$ for each p , so, for each p , there exists $\theta \in (0, 1)$ with

$$\begin{aligned} 0 &= \phi'_p(\theta h) = g'(\theta h) + \frac{p(1-\theta)^{p-1}}{h} g(0) \\ \implies g(0) &= -\frac{hg'(\theta h)}{p(1-\theta)^{p-1}} \end{aligned}$$

But we can see $g(0) = R_{n,f,0}(h)$, so

$$R_{n,f,0}(h) = -\frac{hg'(\theta h)}{p(1-\theta)^{p-1}} = \frac{h^n}{p(n-1)!} (1-\theta)^{n-p} f^{(n)}(\theta h)$$

so putting $p = n$ gives the result. ■

This alternative proof approach will become extremely useful for our next theorem.

Theorem 3.20 (Taylor's theorem: Cauchy remainder)

Let f and $R_{n,f,a}$ be as in [theorem 3.19](#). Then there exists $\theta \in (0, 1)$ such that

$$R_{n,f,a}(h) = h^n \frac{(1-\theta)^{n-1}}{(n-1)!} f^{(n)}(a + \theta h)$$

Proof. This follows immediately from taking $p = 1$ in the final line of the above. ■

Lecture 14 – 24/02/26

Remark. If $f \in C^n([c, d])$ with $[a, a+h] \subset (c, d)$ then, as in previous remark, we can set

$$M_n = \sup_{x \in [a, a+h]} |f^{(n)}(x)|$$

so the theorem gives

$$|R_{n,f,a}(h)| \leq M_n \frac{|h|^n}{(n-1)!}$$

i.e. $R_{n,f,a}(h) = O(h^n)$.

Example

Consider $f(x) = x^q$, $q \in \mathbb{Q}$, which is smooth on $(0, \infty)$ with

$$f^{(n)}(x) = q(q-1) \dots (q-k+1)x^{q-k}$$

so that

$$\begin{aligned} T_{n-1,f,1}(x) &= \sum_{k=0}^{n-1} \frac{q(q-1) \dots (q-k+1)}{k!} x^k \\ &= \sum_{k=0}^{n-1} \binom{q}{k} x^k \end{aligned}$$

and we consider

$$R_{n,f,1}(x) = (1+x)^q - \sum_{k=0}^{n-1} \binom{q}{k} x^k$$

for x in the interval $(-1, 1)$. Let us compare both our remainder options:

- *Lagrange remainder.* We have $\theta \in (0, 1)$ such that

$$R_{n,f,1}(x) = \binom{q}{n} (1+\theta x)^{q-n} x^n$$

Then, if $n \geq q$, $(1+\theta x)^{q-n} = \frac{1}{(1+\theta x)^{n-q}} \leq 1$ for $x \in [0, 1)$, so

$$|R_{n,f,1}(x)| \leq \left| \binom{q}{n} x^n \right| \rightarrow 0$$

as $n \rightarrow \infty$. But this argument fails for $(-1, 0)$.

- *Cauchy remainder.* We have $\theta \in (0, 1)$ such that

$$\begin{aligned} R_{n,f,1}(x) &= (1-\theta)^{n-1} n \binom{q}{n} (1+\theta x)^{q-n} x^n \\ &= (1-\theta)^{n-1} q \binom{q-1}{n-1} (1+\theta x)^{q-n} x^n \\ &= q \binom{q-1}{n-1} \left(\frac{1-\theta}{1+\theta x} \right)^{n-1} (1+\theta x)^{q-1} x^n \end{aligned}$$

Now $1+\theta x = (1-\theta) + \theta(x+1) \geq 1-\theta$, so $\left(\frac{1-\theta}{1+\theta x} \right)^{n-1} \leq 1$ for $n \geq 1$. Hence

$$\begin{aligned} |R_{n,f,1}(x)| &\leq |qx| \left| \binom{q-1}{n-1} \right| |1+\theta x|^{q-1} \\ &\leq C \left| \binom{q-1}{n-1} \right| \end{aligned}$$

by taking the sup over all $x \in (-1, 1)$. But this goes to 0 as $n \rightarrow \infty$. So in fact the Cauchy remainder does give us a stronger result.

Definition 3.21 (Analyticity, non-examinable). We say $f \in C^\infty((c, d))$ is *analytic* if, for every $a \in (c, d)$, there exists an $r > 0$ such that, for any $|h| < r$,

$$f(a + h) = \lim_{n \rightarrow \infty} T_{n,f,a}(h) \iff \lim_{n \rightarrow \infty} R_{n,f,a}(h) = 0$$

in other words, f is equal to its Taylor series.

Example

As we spent the last page showing, x^q is analytic in $(0, 2) \iff (1 + x)^q$ is analytic in $(-1, 1)$.

Remark. Note that, even if $R_{n,f,a}(h)$ does not go to zero, the Taylor series might still converge – just not to the actual function.

§4 Integration of functions

§4.1 Basics

We want to define $\int_a^b f(x) dx$ as the signed area under a graph, which is easy if it's, say, a rectangle. Otherwise, we can try approximating it with slices.

Definition 4.1 (Upper and lower Riemann sums). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Given a partition $\mathcal{P} = \{x_0, \dots, x_n\}$ with $a = x_0 < \dots < x_n = b$ of $[a, b]$, we set

$$L(f, \mathcal{P}) = \sum_{j=1}^n (x_j - x_{j-1}) \inf_{x \in [x_{j-1}, x_j]} f(x)$$

$$U(f, \mathcal{P}) = \sum_{j=1}^n (x_j - x_{j-1}) \sup_{x \in [x_{j-1}, x_j]} f(x)$$

to be the *lower* and *upper Riemann sums* respectively.

Example

Let

$$f : [0, 1] \rightarrow 0, 1$$

$$x \mapsto \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

No matter what $\mathcal{P}$ we take, each of our intervals will contain both a rational and an irrational, so

$$U(f, \mathcal{P}) = 1$$

$$L(f, \mathcal{P}) = 0$$

Remark. For f bounded, i.e. $\sup_{[a,b]} |f| = k$ for some k ,

$$\begin{aligned} U(f, \mathcal{P}) &\leq \sum_{j=1}^n (x_j - x_{j-1}) \sup_{[a,b]} |f| \\ &= k(b - a) \end{aligned}$$

Analogously,

$$\begin{aligned} L(f, \mathcal{P}) &\geq \sum_{j=1}^n (x_j - x_{j-1}) (-\sup_{[a,b]} |f|) \\ &= -k(b - a) \end{aligned}$$

Also, it's clear that $U(f, \mathcal{P}) \geq L(f, \mathcal{P})$. Hence

$$\begin{aligned} \{L(f, \mathcal{P}) : \mathcal{P} \text{ a partition of } [a, b]\} \\ \{U(f, \mathcal{P}) : \mathcal{P} \text{ a partition of } [a, b]\} \end{aligned}$$

are both bounded sets and (clearly) non-empty.

Definition 4.2 (Upper and lower integrals). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. We say

$$\begin{aligned} I_*(f) &= \sup_{\mathcal{P}} L(f, \mathcal{P}) \\ I^*(f) &= \inf_{\mathcal{P}} U(f, \mathcal{P}) \end{aligned}$$

are the *lower* and *upper integrals* respectively.

Definition 4.3 (Riemann integrability). We say f is *Riemann integrable* if $I_* = I^*$. Then we say

$$I(f) = \int_a^b f(x) dx = I_*(f) = I^*(f)$$

Remark. By convention, $\int_a^b f = -\int_b^a f$.

Example

Take f as in the previous example. Then $I_*(f) = 0$ and $I^*(f) = 1$, so f is not Riemann integrable.

Lecture 15 – 26/02/26

The coarsest possible partition for $[a, b]$ is $\{a, b\}$. Our intuition suggests that finer partitions lead to a better estimate for the integral.

Lemma 4.4

Let $\mathcal{P}$ and $\mathcal{P}'$ be partitions of $[a, b]$, and suppose $\mathcal{P} \supseteq \mathcal{P}'$, so $\mathcal{P}'$ is a refinement of $\mathcal{P}$. Then

$$L(f, \mathcal{P}) \leq L(f, \mathcal{P}') \leq U(f, \mathcal{P}') \leq U(f, \mathcal{P})$$

for every bounded $f : [a, b] \rightarrow \mathbb{R}$.

Proof. Let us induct. Take $\mathcal{P} = \{x_0, \dots, x_n\}$ and pick $y \in (x_{k-1}, x_k)$ for some x – then, put $\mathcal{P}' = \mathcal{P} \cup \{y\}$. Then

$$\begin{aligned} & \sup_{x \in [x_{k-1}, y]} f(x), \sup_{x \in [y, x_k]} f(x) \leq \sup_{x \in [x_{k-1}, x_k]} f(x) \\ \implies & (y - x_{k-1}) \sup_{x \in [x_{k-1}, y]} f(x) + (x_k - y) \sup_{x \in [y, x_k]} f(x) \leq (x_k - x_{k-1}) \sup_{x \in [x_{k-1}, x_k]} f(x) \end{aligned}$$

so that $U(f, \mathcal{P}') \leq U(f, \mathcal{P})$ as required. Similar reasoning with the infimum would suffice for the lower sums. Hence we can induct on the elements of $\mathcal{P}' \setminus \mathcal{P}$. ■

Lemma 4.5

Let $\mathcal{P}$ and $\mathcal{P}'$ be any partitions of $[a, b]$. Then

$$L(f, \mathcal{P}) \leq U(f, \mathcal{P}')$$

for every bounded $f : [a, b] \rightarrow \mathbb{R}$.

Proof. Let $\mathcal{P}'' = \mathcal{P} \cup \mathcal{P}'$, so $\mathcal{P}''$ is a refinement of both $\mathcal{P}$ and $\mathcal{P}'$. Then

$$L(f, \mathcal{P}) \leq U(f, \mathcal{P}'') \leq U(f, \mathcal{P}')$$

■

Lemma 4.6

For a bounded $f : [a, b] \rightarrow \mathbb{R}$,

$$I_*(f) \leq I^*(f)$$

Proof. We know that

$$\begin{aligned} & L(f, \mathcal{P}) \leq \inf_{\mathcal{P}'} U(f, \mathcal{P}') \\ \implies & \underbrace{\sup_{\mathcal{P}} L(f, \mathcal{P})}_{I_*(f)} \leq \underbrace{\inf_{\mathcal{P}'} U(f, \mathcal{P}')}_{I^*(f)} \end{aligned}$$

■

Remark. It follows that f is integrable if and only if $I_*(f) \geq I^*(f)$.

Proposition 4.7 (Riemann integrability criterion)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then f is Riemann integrable if and only if, for any $\varepsilon > 0$, there exists a partition $\mathcal{P}$ of $[a, b]$ such that

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$$

Proof. $\Leftarrow$ We have, for any $\varepsilon > 0$,

$$0 \leq I^*(f) - I_*(f) \leq U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$$

$\Rightarrow$ From the definition of sup and inf, we can find partitions $\mathcal{P}$ and $\mathcal{P}'$ such that

$$\begin{aligned} U(f, \mathcal{P}) &\leq I^*(f) + \frac{\varepsilon}{2} \\ L(f, \mathcal{P}') &\geq I_*(f) - \frac{\varepsilon}{2} \end{aligned}$$

which implies $U(f, \mathcal{P}) - L(f, \mathcal{P}') \leq \varepsilon$. To conclude, we need to show this for a single partition. Let $\mathcal{P}'' = \mathcal{P} \cup \mathcal{P}'$, then

$$U(f, \mathcal{P}'') - L(f, \mathcal{P}'') \leq U(f, \mathcal{P}) - L(f, \mathcal{P}') \leq \varepsilon < 2\varepsilon$$

■

Remark. As an exercise, one can show that this is equivalent to a sequential characterisation – finding a sequence of partitions $(\mathcal{P}_n)$ such that

$$U(f, \mathcal{P}_n) - L(f, \mathcal{P}_n) \rightarrow 0$$

as $n \rightarrow \infty$.

Example

Let us attempt to integrate $f(x) = x$ on $[0, 1]$. Let $\mathcal{P}_n = \{0, \frac{1}{n}, \dots, \frac{n-1}{n}, 1\}$. Then

$$\begin{aligned} U(f, \mathcal{P}_n) &= \sum_{j=1}^n \frac{1}{n} \sup_{x \in [\frac{j-1}{n}, \frac{j}{n}]} x \\ &= \frac{1}{n^2} \sum_{j=1}^n \sup_{x \in [j-1, j]} x = \frac{1}{n^2} \sum_{j=1}^n j \\ &= \frac{1}{2} \left(1 + \frac{1}{n} \right) \end{aligned}$$

and

$$\begin{aligned} L(f, \mathcal{P}_n) &= \sum_{j=1}^n \frac{1}{n} \inf_{x \in [\frac{j-1}{n}, \frac{j}{n}]} x \\ &= \frac{1}{n^2} \sum_{j=1}^n \inf_{x \in [j-1, j]} x = \frac{1}{n^2} \sum_{j=1}^n j - 1 \\ &= \frac{1}{2} \left(1 - \frac{1}{n} \right) \end{aligned}$$

Hence

$$\begin{aligned} I_*(f) &= \sup_{\mathcal{P}} L(f, \mathcal{P}) \geq \sup_n L(f, \mathcal{P}_n) = \frac{1}{2} \\ I^*(f) &= \inf_{\mathcal{P}} U(f, \mathcal{P}) \leq \inf_n U(f, \mathcal{P}_n) = \frac{1}{2} \end{aligned}$$

so $I_*(f) \geq I^*(f)$, so f is integrable as required, with $\int_0^1 x = \frac{1}{2}$.

Alternatively, with the Riemann integration criterion,

$$U(f, \mathcal{P}_n) - L(f, \mathcal{P}_n) \leq \frac{1}{n} \rightarrow 0$$

as $n \rightarrow \infty$ (but this doesn't give us the value of the integral).

Example

Let us consider a discontinuous but integrable function, namely

$$f(x) = \begin{cases} 0 & x \leq \frac{1}{2} \\ 1 & x > \frac{1}{2} \end{cases}$$

on $[0, 1]$. Then take $\mathcal{P}_n$ as before for $n \geq 3$. So

$$\sup_{[x_{n-1}, x_n]} f - \inf_{[x_{n-1}, x_n]} f = \begin{cases} 1 & \frac{1}{2} \in [x_{n-1}, x_n] \\ 0 & \text{otherwise} \end{cases}$$

so that

$$U(f, \mathcal{P}_n) - L(f, \mathcal{P}_n) = \frac{1}{n} \rightarrow 0$$

as $n \rightarrow \infty$.

We would like now to generalise the above examples to get whole classes of functions that are integrable.

Proposition 4.8 (Monotone $\implies$ integrable)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded and monotone. Then f is integrable.

Proof. Since f is monotone, f is bounded by $\max(|f(a)|, |f(b)|)$. ■